

Environmental connectivity influences long-term evolutionary outcomes

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Abstract

Evolutionary graph theory uses mathematical representations of networks to model population spatial structure, providing a tractable framework to study how environmental connectivity influences evolution. However, evolutionary graph theory often uses simple genetic representations with limited fitness landscapes and focus on short-term evolutionary outcomes, such as the probability of fixation and average fixation time for new mutations. In this work, we bridge evolutionary graph theory and artificial life approaches to investigate how different patterns of connectivity affect long-term evolutionary adaptation. To do so, we developed an extended evolutionary graph theory platform that supports fully configurable, large fitness landscapes, and we compared evolutionary adaptation on different spatial topologies across three fitness landscapes crafted to isolate different aspects of evolution. We found that, in general, more constrained spatial topologies slow evolutionary adaptation on smooth fitness gradients but can facilitate improved fitness landscape exploration and valley crossing. Next, we compared the evolution of complex adaptive traits in populations of self-replicating computer programs (digital organisms) evolving on different topologies using Avida. In populations of digital organisms, we found variation in the number of complex traits and reproductive strategies evolved across different patterns of connectivity. For example, some structures favored organisms with smaller genomes capable of faster self-replication, while others favored more complex organisms. More broadly, this work demonstrates a path forward for extending evolutionary graph theory to study long-term evolutionary outcomes and poses critical considerations for how we study the effects of spatial structure in artificial life systems.

Introduction

The environment's spatial structure dramatically influences evolutionary dynamics (Kool et al., 2013; Cramer et al., 2023). At short time scales, spatial structure can influence the rate of evolutionary adaptation by either amplifying or suppressing the fixation of new mutations (Lieberman et al., 2005; Kuo and Carja, 2024a; Chakraborty et al., 2023). At longer time scales, the environment's connectivity affects the evolution and maintenance of biodiversity (Hernández-Hernández et al., 2021; Schluter and Pennell, 2017; Stein et al., 2014). However, empirical investigations of how

different patterns of connectivity impact evolutionary outcomes can be challenging to conduct using natural organisms, as they require precise control over spatial connectivity and the ability to observe evolution for many generations.

Evolutionary graph theory is a mathematical framework for studying how population structure affects evolution (Lieberman et al., 2005). An evolutionary graph defines the connectivity of a set of nodes, each of which can be occupied by a single individual¹. Edges in the graph represent the spatial pattern of interaction in the population. Evolutionary graph theory affords full experimental control over a population's spatial connectivity, while remaining mathematically tractable for small graphs. Indeed, evolutionary graph theory has shown that spatial connectivity influences many short-term dynamics, such as the likelihood that a new mutant's lineage sweeps the population (i.e., fixation probability) and the mean time to fixation for new mutations (Allen et al., 2021; Möller et al., 2019; Kuo and Carja, 2024a).

However, short-term evolutionary dynamics may not predict longer-term outcomes. Conventional evolutionary graph theory frameworks use limited fitness landscapes (i.e., mappings from genotype to fitness), and individuals have simple genetic representations, often comprising a numeric or categorical value with just a few possible variants. Many artificial life systems overcome these limitations, featuring populations of "digital organisms" with complex genetic representations that evolve in environments with non-trivial fitness landscapes (Ofria et al., 2009; Batut et al., 2013; Bohm et al., 2017; Dolson and Ofria, 2021). While more complex and less mathematically tractable, artificial life systems afford the same level of control over a population's spatial connectivity as in evolutionary graph theory frameworks. Indeed, artificial life studies have investigated how environmental connectivity influences longer-term evolutionary dynamics (e.g., Dolson et al., 2017; Vostinar and Ofria, 2019; Misevic et al., 2015). However, many of these studies focus on a narrow set of possible environmental connectivity patterns, often limited to comparing populations in

¹Nodes are often considered to be occupied either by a single individual or by a genetically homogeneous subpopulation.

“well-mixed” (fully connected) environments with those on a two-dimensional lattice.

In this work, we bridge evolutionary graph theory and artificial life approaches to studying the effects of spatial structure on evolution. First, we use an extended evolutionary graph theory system to investigate evolution on different spatial topologies in the context of three different fitness landscapes, each designed to isolate different characteristics of evolution: exploiting a smooth fitness gradient, exploring multiple pathways in a fitness landscape, and crossing fitness valleys. Then, we use the Avida Digital Evolution Platform (Ofria et al., 2009) to compare the evolution of complex traits in populations of self-replicating computer programs across different patterns of spatial connectivity.

Across all experiments, we observe that spatial structure influences evolutionary outcomes, and different patterns of spatial connectivity have different effects depending on the fitness landscape and study system. With our evolutionary graph theory study system, we find that more constrained graph topologies (i.e., graphs with regions of low connectivity) tend to increase an evolving population’s capacity for multi-path exploration and valley crossing but slow down the exploitation of smooth fitness gradients. In our Avida experiments, we also observe differences in the evolution of complex adaptive traits and genome length in populations evolved on different spatial topologies. Because many artificial life systems use lattice-structured populations, we additionally test how choice of lattice shape (i.e., height and width) affect evolutionary outcomes. Across both study systems, we show that choice of lattice shape can significantly alter evolutionary outcomes.

Background

Environmental connectivity mediates the range of possible interactions within a population as well as between species. In natural systems, connectivity is often determined by spatial distance and other abiotic and biotic factors, such as winds, ocean currents, physical barriers, or the migratory patterns of other species (Moritz et al., 2013). The environment’s connectivity profoundly affects eco-evolutionary dynamics at all time scales, such as the exchange of individuals between populations (i.e., gene flow) and extinction risk by determining where species can move in response to environmental change (Baguette et al., 2013).

Within a single population, environmental connectivity influences the strength of selection (Allen et al., 2020), genetic drift (Willi and Mänttinen, 2011; Schneider et al., 2016), fixation probability and time for new mutations (Allen et al., 2021), and the overall rate of evolutionary adaptation (Perfeito et al., 2008; Bailey and Kassen, 2012; Kuo and Carja, 2024b). Evolutionary graph theory uses tools from network theory to model evolution on a broad range of different graph structures (Möller et al., 2019). Indeed, evolutionary graph theory studies have shown that the

particular pattern of connectivity influences short-term evolution, and connectivity can be manipulated to directly influence those outcomes (Kuo and Carja, 2024a). Additionally, a given structure’s effects can change depending on factors like mutation rate, the fitness landscape, and the relative fitness of individuals in the population.

Artificial life systems have also investigated the impact of spatial structure on evolutionary processes. These systems have allowed investigations of a broad set of evolutionary outcomes, as they typically feature richer virtual environments and more complex genetic representations as compared to those used in evolutionary graph theory. For example, spatial resource heterogeneity has been shown to increase diversity and promote the evolution of complex traits in populations of digital organisms (Dolson et al., 2017). Digital populations evolving on a lattice structure have been shown to maintain greater phylogenetic diversity than those evolving in a well-mixed environment (Moreno et al., 2025). Spatial structure has also been shown to both increase or decrease the evolution of mutualisms relative to well-mixed environments under different circumstances (Vostinar and Ofria, 2019; Doebeli and Knowlton, 1998; Akçay, 2015).

Many of these digital evolution studies compare populations on a lattice structure with those in a well-mixed (fully connected) environment. Few digital evolution studies have focused on the particular pattern of environmental connectivity. Misevic et al. 2015 used the Aevol platform to compare how two different lattice shapes affected the evolution and maintenance of cooperation in populations of digital organisms, finding that populations in square, 100 by 100 toroidal lattices evolved more cooperation than populations in flatter 2 by 2500 lattices. If varying lattice shape can have strong effects on the evolution of a complex cooperative behavior, we expect other connectivity patterns to impact a broad range of evolutionary outcomes.

Methods

We assess the role of environmental connectivity using two model systems. First, we use an extension of conventional evolutionary graph theory frameworks to investigate how different spatial topologies influence evolution across three fitness landscapes. Next, we investigate whether choice of spatial topology influences the evolution of complex traits with Avida (Ofria et al., 2009).

Evolutionary graph theory system

We developed a simple computational model that extends conventional evolutionary graph theory frameworks. The environment is defined by a graph where nodes represent locations that can be occupied by a single individual, and edges between nodes define connectivity between locations. Like many evolutionary graph theory frameworks, “organisms” comprise a single numeric value (their genome). Our

system supports the study of long-term evolutionary dynamics by allowing users to configure arbitrarily large fitness landscapes that specify the mutational adjacency of different genotypes as well as each genotype's corresponding fitness.

Each update corresponds to a single generation of evolution. Generational turnover follows a death-birth procedure. For each location, we select a parent from a neighboring location to asexually reproduce into that location. Parents are selected using fitness proportionate (or roulette) selection (Goldberg, 1989) where individuals are probabilistically chosen proportionally to their fitness. Fitness is determined according to the treatment-specific fitness landscape (each described below) that defines the mapping from an individual's genotype to their fitness value. Each update, we first choose the parents that will reproduce into each location. We then carry out reproductions simultaneously, ensuring that the order of parent selection does not matter. When a parent reproduces, there is a 1.25% chance the offspring mutates to a random, mutationally adjacent genotype. We limited experiments with this model to 100,000 generations.

Fitness landscapes Inspired by the diagnostic landscapes from (Hernandez et al., 2022), we constructed three fitness landscapes to assess spatial connectivity's effect on different aspects of evolution: evolutionary exploitation, exploration of multiple paths, and crossing fitness valleys.

Smooth gradient landscape: Populations start at the bottom of a smooth fitness gradient that leads to the global fitness peak. The gradient comprises 1,000 mutational steps, and each step corresponds to a roughly 3.5% fitness increase.

Multi-path landscape: The multi-path landscape comprises 100 initially identical gradients, or "paths", to local fitness peaks. Each path features an identical upward slope of a $\sim 3.5\%$ fitness increase per mutational step. Path lengths are unique, ranging from 1 to 100 mutational steps. The length of a path defines its fitness potential, so the longest path leads to the global optimum. Individuals in the initial population are equally likely to mutate onto any of the 100 gradients, but individuals cannot mutate from one path to another, except at the lowest (starting) point in the fitness landscape. To consistently reach the global fitness peak, populations must be able to explore different pathways in the fitness landscape until lineages on the longest path eventually outcompete those on shorter paths. This is most easily achieved if multiple subpopulations can simultaneously explore different pathways until the subpopulation on the optimal pathway eventually outcompetes all other subpopulations.

Valley-crossing landscape: The valley-crossing landscape features 100 repeating fitness valleys, similar to those used in (Ferguson et al., 2024). To cross a valley, a lineage must accumulate four deleterious (each a 5% fitness penalty) and then one beneficial mutation. Each fitness peak corresponds to a 50% fitness benefit over the previous peak. Achieving

high fitness requires populations to "explore" across deleterious valleys in order to reach the next peak.

We include implementation and configuration details for each fitness landscape in supplemental material (Gordon et al., 2025b).

The Avida Digital Evolution Platform

For brevity, we direct readers to (Ofria et al., 2009) for a detailed description of Avida. Briefly, Avida instantiates populations of self-replicating computer programs (digital organisms) that compete for space in a finite virtual environment. Digital organisms reproduce asexually by executing `copy` operations to copy their genome instruction-by-instruction and then dividing with a `divide` operation. The `copy` and `divide` operations are imperfect and can result in mutated offspring. In this work, the `copy` operation errs with a probability of 0.25% per instruction copied, potentially resulting in single-instruction substitution mutations. The `divide` operation results in single-instruction insertions and deletions at a per-division probability of 5%. We chose mutation rates commonly used in Avida studies (Lenski et al., 2003).

When an organism reproduces, its offspring is placed in a different neighboring location, replacing that location's previous occupant. By default in Avida, organisms improve their rate of replication by performing designated Boolean logic functions to increase their genome expression rate. In preliminary experiments, we found that variation in genome expression produced degenerate behavior in spatial topologies with regions of low connectivity, creating regions where organisms were "frozen in time" (see supplement for more details (Gordon et al., 2025b)). We deviate from this default Avida configuration to avoid degenerate model behavior and to align more closely with the behavior of conventional evolutionary graph theory models (e.g., Kuo and Carja, 2024a).

In this work, organisms express their genome at a fixed rate (in contrast to Avida's default setting). An offspring replaces the neighboring organism with the largest age divided by "merit" where an organism's merit is determined by the number of logic functions performed. That is, older organisms that have performed fewer logic functions are preferentially replaced; ties are broken randomly.

For Avida experiments, we initialize all locations in the environment with the default Avida ancestor capable of only self-replication, and we evolved populations for 400,000 updates ($\sim 48,000$ generations under well-mixed conditions, Gordon et al. 2025b). A single update corresponds to each organism in the population expressing 30 instructions in their genome. We use the Logic-77 environment, wherein organisms can increase their merit by performing any of 77 distinct 1-, 2-, and 3-input logic operations (Ofria et al., 2009). In this work, all logic operations provide equal benefit of doubling the organism's merit value, which reduces their chance of replacement by the offspring of a neighboring organism (potentially increasing their reproductive fit-

ness). We provide Avida configuration files used for these experiments in our supplemental material (Gordon et al., 2025b); except for the deviations from defaults documented here, we kept defaults for all other Avida configuration values (Bryson et al., 2021).

Experimental design

We performed a total of eight independent experiments, six with our evolutionary graph theory system and two with Avida. For all experiments, we limited the number of locations in the environment (and population size) to 3,600, matching the most commonly used population size in Avida studies (e.g., Lenski et al., 2003; Dolson and Ofria, 2017; Lalejini et al., 2021). For each experiment, we varied the environment’s connectivity across treatments.

Each experiment featured one of two sets of graph structures used to define environmental connectivity. In our “**varied graph structure**” experiments, we included ten graph structures commonly used in evolutionary graph theory Möller et al. (2019), each described below.

In our “**lattice shape**” experiments, we focused on the effect of varying toroidal lattice shape as in (Misevic et al., 2015). We compared evolution on lattice shapes that varied how vertices were distributed across the height and width of the lattice. In experiments with our evolutionary graph theory system, we included seven lattice configurations (specified as height x width): 60x60 (square), 30x120, 15x240, 4x900, 3x1200, 2x1800, and 1x3600 (linear cycle). In our Avida experiment with varied lattice shape, we included fewer configurations: 60x60, 15x240, 2x1800, and 1x3600. We did so based on results from preliminary experiments where intermediate lattice configurations behaved similarly (see supplement Gordon et al. 2025b), and the large number of multiple comparisons reduced our statistical power to differentiate between treatment effects. For all experiments, we ran 50 independent replicates of each treatment.

Spatial topologies Brief descriptions of graph structures used to define environmental connectivity are given below:

- **Fully connected (well-mixed):** All vertices are connected to all other vertices. This is commonly referred to a “well-mixed” environment in digital evolution studies.
- **Comet-kite:** A fully connected set of core vertices with additional random “tails” (Pavlogiannis et al., 2017). To generate a comet-kite graph, we construct a fully connected core and attach t initial tail vertices to a single vertex in the core. Then, we sequentially attach additional tail nodes to existing tail nodes (chosen uniformly at random). In this work, we used a core size of 1440 (40% of the graph), 720 initial tail vertices (20% of the graph), and 1440 additional tail vertices (40% of the graph). These parameters were chosen based on (Möller et al., 2019).
- **Windmill:** Contains n size- k cliques that each share a single hub vertex. In this work, $n = 60$ and $k = 59$,

resulting in a graph with 3,540 vertices. To increase the size of the graph to 3,600 vertices, we attached 60 additional vertices to randomly chosen vertices in the Windmill structure; this random attachment did not change the overall structure the graph (i.e., did not add additional connections between cliques).

- **Clique ring:** Contains l size- k cliques, each joined by length- c linear chains into a ring. For this work, $l = 115$, $k = 30$, and $c = 5$. That is, 30 size-115 cliques are arranged in a ring, each connected by length-5 linear chains.
- **Linear chain:** An acyclic linear chain where each vertex is connected to its two neighbors.
- **Cycle:** A linear chain, but the vertices at the two ends of the chain are connected, forming a cycle.
- **Toroidal Lattice:** Vertices are organized into a toroidal grid with edges connecting each vertex to four neighboring vertices. The vertices in the top and bottom rows and the leftmost and rightmost columns are considered neighbors, respectively. The configured width and height of the lattice structure varied by treatment.
- **Random Waxman:** A graph generated by placing vertices uniformly at random in a 2-dimensional space (Waxman, 1988). Each pair of vertices distance d from one another are connected with probability $p = \beta e^{-d/\alpha L}$. Here, each dimension ranged from 0 to 1, $\beta = 0.4$, and $\alpha = 0.2$.
- **Star:** A graph with one internal vertex, and all other vertices are connected exclusively to the internal vertex.
- **Wheel:** A cycle where all vertices in the cycle are connected to a single “hub” vertex.

For graphs generated by stochastic graph generation algorithms, we generated a different graph for each replicate using that structure. We used the networkx Python library (Hagberg et al., 2008) to generate the star, windmill, cycle, wheel, and random Waxman graphs. All graph generation code and all graphs used in our experiments are included in supplemental material (Gordon et al., 2025b).

Statistical Analyses

When comparing distributions of measurements taken from independent treatments within a single experiment, we screened for statistical differences in distributions using a Kruskal-Wallis test (Kruskal and Wallis, 1952). For comparisons in which the Kruskal-Wallis test was significant ($p < 0.05$), we performed Wilcoxon rank-sum tests (Wilcoxon, 1992) to identify pairwise differences. All reported statistics are corrected for multiple comparisons using the Holm-Bonferroni method as appropriate (Chen et al., 2017).

Code and data availability

Our evolutionary graph theory system is implemented in C++ using the MABE2 platform (<https://github.com/mercere99/MABE2>). All configuration files, experiment software, data analyses, and statistical results can

be found in supplemental material hosted on GitHub and archived with Zenodo (Gordon et al., 2025b). Experiment data are archived on the Open Science Framework at (Gordon et al., 2025a).

Results and Discussion

Spatial connectivity influences evolutionary adaptation across different fitness landscapes

First, we used our evolutionary graph theory system to investigate the effect of spatial topology in the context of three different fitness landscapes: a smooth gradient, a multi-path exploration, and a valley-crossing landscape. Figures 1 and 2 show results on each of these three landscapes for the varied graph structure and lattice shape experiments, respectively. Across all fitness landscapes, we observed significant effects of spatial topology. However, the effect of any particular topology varied by fitness landscape.

Spatial constraints slow evolutionary adaptation on smooth fitness gradients On the smooth gradient fitness landscape, we compared the number of generations elapsed before a maximum fitness (50) individual evolved (Figures 1A and 2A). Across both varied graph and lattice shape experiments, populations evolved on only the star and wheel spatial topologies failed to produce maximum fitness individuals within 100,000 generations. For each replicate that failed to reach the global optimum, we projected the expected generation for a maximum fitness individual to evolve using the greatest fitness achieved after 100,000 generations (assuming a fixed rate of subsequent adaptation).

In the varied graph experiment (Figure 1), the well-mixed regime imposes no spatial constraints on the population. Overall, we found that imposing spatial constraints on a population increased the amount of time for a maximum-fitness individual to evolve. The well-mixed regime reached maximum fitness in fewer updates than all other topologies except the random Waxman treatment (corrected Wilcoxon rank-sum tests, $p < 10^{-4}$). We failed to detect significant differences between linear chain and cycle regimes, indicating that the asymmetry of the linear chain as compared to the cycle graph has no strong effect on evolutionary adaptation for the smooth gradient landscape. Differences in generations to maximum fitness between all other pairs of graph structures were statistically significant ($p < 10^{-4}$), ordered from fastest to slowest to maximum fitness: well-mixed and random Waxman, comet-kite, windmill, toroidal lattice, clique ring, cycle and linear chain, wheel, and star.

The star graph's topology caused degenerate model behavior, impeding evolutionary adaptation. The star graph comprises 3,599 "outer" locations, each with each a single connection to a central "hub" location. Every generation, just two individuals in the star regime reproduce, bottlenecking the population. The individual occupying the hub location reproduces into all outer locations, and one of the 3,599

outer location occupants is chosen by roulette selection to reproduce into hub location. All outer location individuals have similar fitness because of repeated bottlenecking, so any one individual's probability of selection by roulette selection is low, causing a greater degree of genetic drift. We expect that populations evolving on the wheel graph also experience this dynamic, but in a less extreme manner because of connections between outer nodes in the wheel graph.

Varying lattice shape more clearly shows that increasing spatial constraints on a population decreases the rate of evolutionary adaptation on a smooth fitness gradient (Figure 2A). We detected statistically significant differences in generations to maximum fitness between all pairs of lattice shapes ($p < 10^{-3}$). For all comparisons, lattice shapes more evenly distributed in length and height produced maximum fitness individuals faster than lattices with more elongated shapes.

In the varied graph structure experiment, differences in topology alter selection in two ways. First, topologies with more neighbors per individual lower the probability of any single individual reproducing, even those with beneficial mutations. This lowers the fixation probability of a beneficial mutation, slowing the rate of adaptation. Second, once a beneficial mutation begins to sweep, it should sweep faster in more densely connected populations than in more spatially constrained populations, increasing the rate of adaptive evolution. In our lattice shape experiments, all locations in all lattices other than 1x3600 and 2x1800 have four neighboring locations, making average fixation time the only difference between treatments.

The tradeoff of fixation probability and fixation time is well known from evolutionary graph theory (Tkadlec et al., 2019). Our experiments use a high enough mutation rate that multiple offspring are mutated every generation. As such, even if weak selection pressure prevents some beneficial mutations from reaching fixation in high density graphs, the inflow of new, beneficial mutations is sufficiently high to provide many opportunities for beneficial mutations to sweep. Our findings on the smooth gradient landscape largely align with predictions from evolutionary graph theory, except on graph structures where increasing the graph's size causes degenerate behavior (e.g., the star and wheel graphs). Prior evolutionary graph theory research has not observed this degenerating behavior, generally finding that the star graph strongly amplifies adaptive evolution (Lieberman et al., 2005; Möller et al., 2019), likely due to a focus on small, mathematically tractable graphs. Our data highlight the impact of graph size for these dynamics.

Constrained spatial topologies facilitate fitness landscape exploration We constructed the multi-path and valley-crossing fitness landscapes to measure how different spatial topologies affect evolutionary exploration. Figures 1B and 1C show results from varied graph structure populations on the multi-path and valley-crossing land-

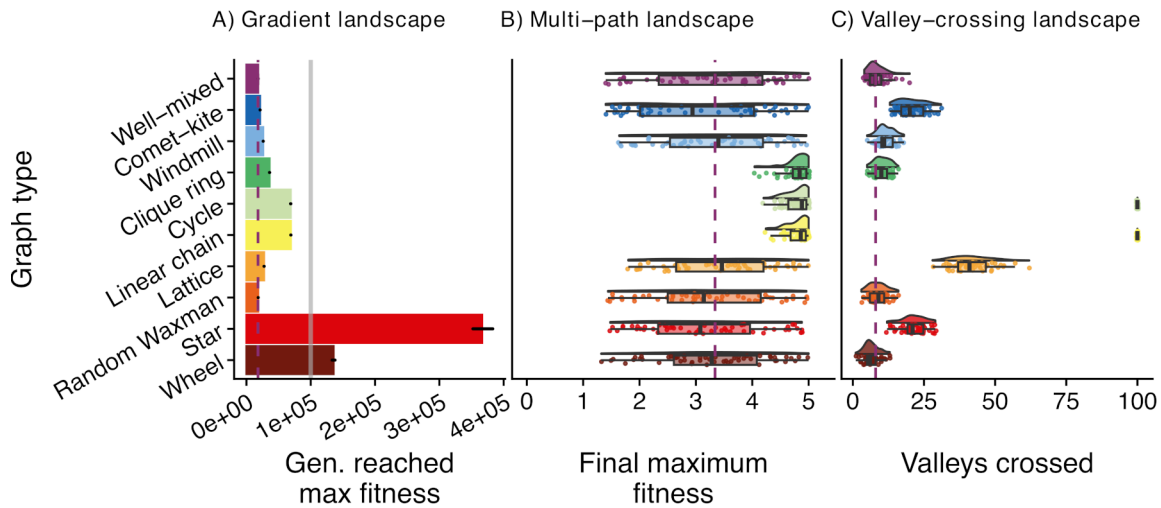


Figure 1: **Varied graph structure results** after 100,000 generations on A) smooth gradient landscape (generation reached global optimum), B) multi-path landscape (final maximum fitness), and C) valley-crossing landscape (number of valleys crossed). In each panel, the dashed vertical line corresponds to the median measurement from the well-mixed treatment. In panel A, the gray vertical line indicates the number of generations elapsed during the experiment, values beyond the gray line are projected estimations, and the black line on the end of each bar indicates a bootstrapped 99% confidence interval. Kruskal-Wallis tests for each fitness landscape were statistically significant ($p < 10^{-4}$).

scapes, respectively. On the multi-path landscape, populations on the clique ring, cycle, and linear chain structures achieved significantly greater final fitness values than all other graph structures ($p < 10^{-4}$). We failed to detect significant differences between all other treatment pairs. These data show that imposing spatial constraints does not universally promote consistent multi-path exploration, as outcomes for many of the topologies were indistinguishable from well-mixed.

The clique ring, cycle, and linear chain topologies were the only structures in our experiment that contain isolated chain substructures. Chain structures impose strong constraints on how quickly offspring from lineages originating in different regions of the environment can compete with one another. For example, in a linear chain, offspring of a lineage originating on one end of the chain would not compete with descendants of organisms from the opposite end of the chain for *at least* 1,800 generations. This reduces the rate of selective sweeps, and allows for multiple sub-populations to co-exist exploring different regions of the fitness landscape. Indeed, we observed greater phylogenetic diversity in populations evolving in clique ring, cycle, and linear chain regimes (see Gordon et al. 2025b). On average, we observed coalescence events in other treatment populations by generation 1,000, which resulted in those populations getting “stuck” on the pathway occupied by the first lineage that swept.

On the valley-crossing fitness landscape, all populations evolved on the chain and cycle graphs crossed all 100 fitness valleys (Figure 1C). With the exception of the wheel and random Waxman graphs, adding spatial constraints improved valley crossing relative to a well-mixed environment ($p < 0.05$). Broadly, graphs with small diameters (greatest

distance between two vertices) and graphs comprising large cliques crossed the fewest valleys.

Figures 2B and 2C show results from our varied lattice shape experiments on the multi-path and valley-crossing landscapes, respectively. Populations evolved on more elongated lattices achieved greater fitness on the multi-path landscape and crossed more valleys in the valley-crossing landscape. That is, more spatially constrained lattice shapes promoted more fitness landscape exploration relative to less spatially constrained shapes. For example, the 1x3600, 2x1800, 3x1200, and 4x900 lattice regimes achieved significantly greater final fitness on the multi-path landscape than all other lattice regimes ($p < 0.04$). Populations in more elongated lattice regimes also crossed the greatest number of valleys. For example, 1x3600 and 2x1800 regimes crossed all 100 fitness valleys, and the 1x3600, 2x1800, 3x1200, and 4x900 lattice regimes crossed significantly more valleys than all other lattice regimes ($p < 10^{-4}$).

These results largely align with expectations when considering the tradeoff between fixation time and fixation probability across different spatial topologies. More constrained topologies with lower fixation probabilities and longer average fixation time provide more time for subpopulations to coexist. On the multi-path landscape, this allows for subpopulations to explore different pathways, increasing the likelihood that subpopulations on longer paths outcompete those on shorter paths. On the valley-crossing landscape, populations on densely connected structures with faster fixation times experience greater purifying selection, preventing them from maintaining lineages that have begun to accumulate the deleterious mutations necessary to cross the next valley. Our results also align with previous work studying val-

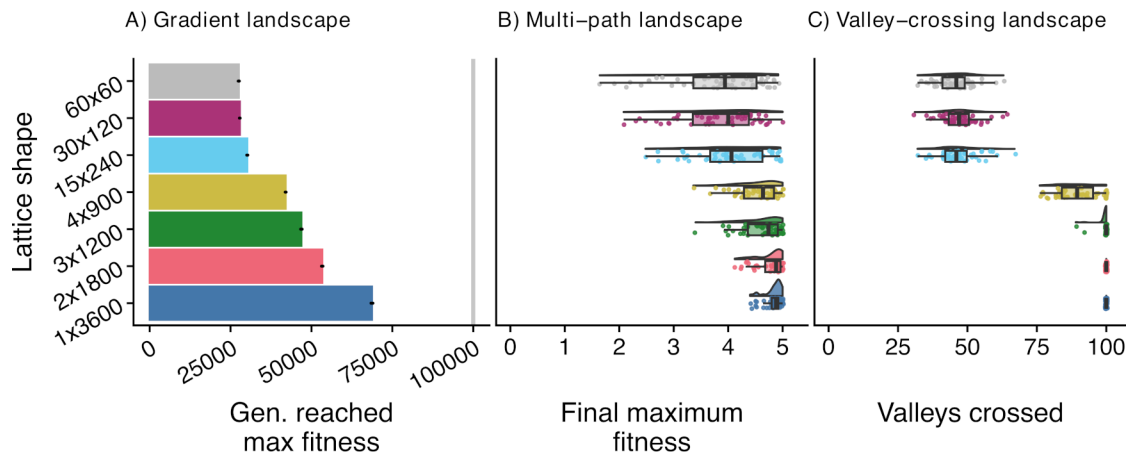


Figure 2: **Varied lattice shape results** after 100,000 generations of evolution on A) smooth gradient landscape (generation reached global optimum), B) multi-path landscape (final maximum fitness), and C) Valley-crossing landscape (number of valleys crossed). In panel A, the gray vertical line indicates the number of generations elapsed during the experiment, and the black line on the end of each bar indicates a bootstrapped 99% confidence interval. Kruskal-Wallis tests for each fitness landscape were statistically significant ($p < 10^{-4}$).

ley crossing via adaptive momentum (Ferguson et al., 2024; Bohm et al., 2024). Adaptive momentum argues that the fitness advantage of progeny from a lineage that has recently crossed one fitness valley may allow for the accumulation of enough deleterious mutations to cross the next valley. Indeed, Ferguson et al. 2024 demonstrate this dynamic on linear chain spatial structures, as they maximize the fixation time of beneficial mutations.

Our findings also align with evolutionary graph theory on smaller fitness landscapes. For example, in a two-step landscape with a fitness valley, Kuo and Carja 2024b found that lattice and detour graphs generally exhibited better rates of valley crossing than other graph structures in their study (size-100 graphs, and chain / cycle graphs were not tested). Notably, a detour graph contains a linear chain with both ends connected into a fully connected core.

Spatial topology influences the evolution of complex adaptive features

Next, we investigated the evolution of complex traits on different spatial topologies using Avida. Figures 3A and 3B show the number of distinct logic functions expressed by populations at the end of the experiment for the varied graph structure and lattice shape experiments, respectively.

In the varied graph structure experiment, six spatial topologies promoted the evolution of a greater number of Boolean logic functions relative to the well-mixed regime (corrected Wilcoxon rank-sum tests, $p < 0.04$): windmill, toroidal lattice, clique ring, linear chain, and wheel. Windmill populations performed the greatest number of tasks relative to all other topologies ($p < 0.04$).

We failed to detect differences in population-level function evolution between well-mixed and the cycle, random Waxman, and comet-kite graph structures. The star struc-

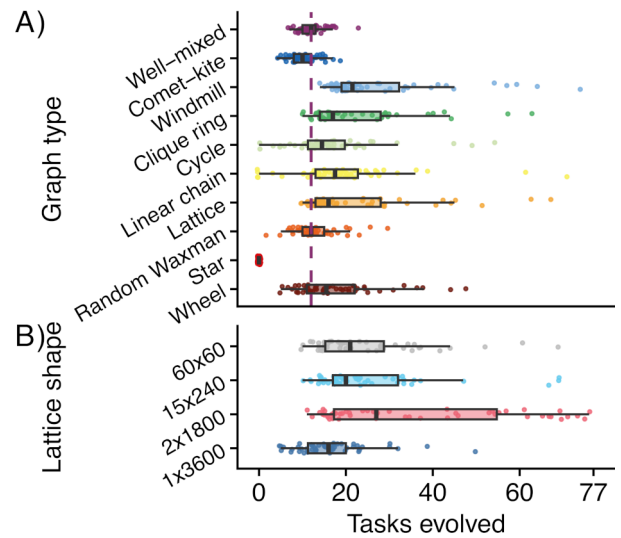


Figure 3: **Number of Boolean logic functions evolved** across A) varied graph structures and across B) varied lattice shapes. The vertical dashed line in panel A indicates the median number of functions evolved in the well-mixed treatment. Kruskal-Wallis tests for each panel comparison were statistically significant ($p < 10^{-4}$).

ture suppressed the evolution of logic functions relative to well-mixed ($p < 10^{-4}$), as it elicited degenerate behavior in Avida. As in our simpler model system, all individuals in outer locations reproduce into the central hub location, bombarding the central location with offspring. In Avida, digital organisms must copy themselves to reproduce, which takes time, so any organism born into the center location failed to copy itself prior to being replaced by the offspring from an outer location. This dynamic isolated all outer locations from one another, eliminating competition between them.

Population-level function completion does not necessarily

mean that the population contains a dominant (most numerous) genotype that performs many functions. We observed variance in the number of distinct functions completed by dominant organisms as well as the proportion of distinct functions completed by organisms of non-dominant genotypes (see Gordon et al. 2025b). For example, the windmill graph's promotion of population-level function evolution appears to be driven by increased diversity. Averaged across windmill replicates, nearly 40% of distinct logic functions were performed by organisms with non-dominant genotypes, which accounts for the differences between it and other spatial topologies that amplified function evolution.

We found that some spatial topologies promoted more “fast-replicator” strategies than others. For example, dominant genotypes from the wheel, linear-chain, and cycle structures evolved the shortest genome lengths, and had the fastest replication times (see Gordon et al. 2025b). The dominant genotypes across comet-kite populations performed a median of 0 logic functions (mean of 0.42). Indeed, 95% of functions completed by comet-kite populations (average of 10.3) were completed by organisms with a non-dominant genotype, which contrasts with other spatial topologies that range from an average of 4.32% (toroidal lattice) to 39.8% (windmill). This indicates multiple, co-existing strategies present in comet-kite populations where one subpopulation specializes as fast replicators that perform few logic functions, and another subpopulation specializes on expressing logic functions at the cost of replicate rate.

Given the dynamics observed on other graph structures, the comet-kite data becomes more intuitive. The comet-kite graph's topology creates two types of regions joined by a single connection: a fully connected “core” region and a “tail” region created by sequentially attaching a new vertex to a random other tail vertex. In this work, the fully connected core contains 40% of the locations, and the tail region contains 60% of locations. Birth count data verified that the location joining the two regions of the graph structure undergoes bombardment (similar to that of the star graph), preventing gene flow through that location. Locations in the core are dominated by organisms performing logic functions as defense against replacement, and fast replicators that perform very few functions dominate the tail region.

We also observed differences in evolutionary outcomes as a result of varying the toroidal lattice shape (Figure 3B). The 2x1800 lattice populations evolved to perform more logic functions than in any of the three other configurations ($p < 0.04$), and 1x3600 populations evolved to perform the least number of functions than in any other configuration ($p < 0.003$). We failed to detect significant differences in functions performed between 60x60 and 15x240 lattices.

As in the varied graph Avida experiment, some differences in population-level function performance were driven by diversity in the population. For example, in the 1x3600 populations, an average (across replicates) of 27.6% of

population-level functions were performed by organisms with non-dominant genotypes. In 60x60 and 15x240 populations, just under 5% of population functions were performed by organisms with non-dominant genotypes. On average, dominant genotypes in 2x1800 performed the greatest number of logic functions, but this difference was not significant when compared to the 15x240 and 60x60 regimes. Indeed, a greater proportion of population-level functions were performed by organisms with non-dominant genotypes in 2x1800 than in the 15x240 and 60x60 lattices.

Conclusion

Our results must be taken in context of the many limitations of our study. For example, population size, mutation rate, and the magnitude of fitness effects from mutations are known to influence the effect of different spatial structures on evolutionary adaptation (Kuo and Carja, 2024b; Tkadlec et al., 2019). Future work should vary mutation rate and explore fitness landscapes with different distributions of mutational effects. For experimental tractability, we limited ourselves to one parameterization of each graph type. However, graph parameterizations can be tuned to influence evolution in different ways (Kuo and Carja, 2024a; Möller et al., 2019), motivating future studies focused on varying graph parameterizations on a narrower set of graph types.

This work demonstrates how we can combine approaches from evolutionary graph theory and artificial life to study how environmental connectivity influences long-term evolutionary outcomes. Our results also pose critical considerations for how we study the effects of spatial structure in evolutionary graph theory and in artificial life studies. Our data show that the size of some graph types can strongly influence their effects; yet, most evolutionary graph theory focuses on small graph sizes. For example, scaling the star graph to 3,600 vertices produced degenerate behavior in our study systems. Indeed, some of our experimental observations on certain spatial topologies (e.g., “bombardment” in the star graph) have few analogues in natural systems, as they were driven by eccentricities of our model systems. However, we still include these results, as similar issues could arise in other computational models or in laboratory experiments imposing similar spatial topologies.

Many previous digital evolution studies determine the effect of spatial structure by comparing populations evolved on a square lattice with well-mixed populations (e.g., Vostinar and Ofria, 2019; Moreno et al., 2025; Dolson et al., 2017). Our data highlight that the particular pattern of connectivity, including lattice shape, influences evolution. These results motivate future studies of how different spatial topologies affect a broad range of dynamics. By connecting those studies back to predictions from evolutionary graph theory, the artificial life community is uniquely positioned to bridge the knowledge gap in our understanding of spatial structure's effects beyond short-term outcomes.

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